Des. Codes Cryptogr. (2015) 77:713–724
DOI 10.1007/s10623-015-0128-9

Power map permutations and the discrete log problem

Peter Landrock[1]

Received: 15 October 2014 / Revised: 30 July 2015 / Accepted: 30 July 2015 /
Published online: 26 August 2015
© Springer Science+Business Media New York 2015

Abstract For g any generator of the multiplicative subgroup of the integers modulo a prime p, we consider the permutation that maps i to g^i, for $1 \leq i \leq p-1$, the inverse of the discrete log map. Such a permutation is the composite of cycles, and if a cycle is of length n and starts with g^s, the last element of the cycle is s, so we can solve the DLP in at most n steps. We characterise all cycles of length 1 (fixpoints) as well as the number of generators for which an element is a fixpoint. In addition, we identify a number of conditions under which there will be orbits of length 2, 3 and 4, and provide a simple formula for switching between generators. Short orbits would all obviously provide keys that would be disastrous in public key schemes such as DSA and Diffie-Hellman and thus should be avoided. This is emerging research, and it remains to be seen if such keys can be avoided by merely using good random bit generators, or whether certain primes and generators are inherently weak.

Keywords The discrete log problem · Fixpoints · Short orbits · Weak keys for Diffie-Hellman

Mathematics Subject Classification 11T71

1 Power map permutations

Let p be an odd prime and g a generator of the multiplicative group modulo p. Consider the isomorphism $i \mapsto g^i \bmod p$ between the additive group $(\mathbb{Z}/(p-1)\mathbb{Z}, +)$ and the multiplicative group $\mathbb{Z}_p^* = (\mathbb{Z}/p\mathbb{Z}^*, \cdot)$. This induces a bijective map μ_g on $\mathbb{Z}_p^*$ which obviously is *not* a homomorphism, but merely a permutation. Indeed, just think of the elements in

This is one of several papers published in *Designs, Codes and Cryptography* comprising the "Special Issue on Cryptography, Codes, Designs and Finite Fields: In Memory of Scott A. Vanstone".

✉ Peter Landrock
landrock@cryptomathic.com

[1] Cryptomathic Ltd., Cambridge Science Park, Cambridge CB4 0WG, UK

$(\mathbb{Z}/(p-1)\mathbb{Z}, +)$ as $\{1, 2, \ldots, p-1\}$—which are precisely the elements of $\mathbb{Z}_p^*$—rather than $\{0, 1, 2, \ldots, p-2\}$, simply choosing $p-1$ whenever 0 occurs. We shall refer to these maps as *power maps*. And one more thing: Even though it is a permutation on the multiplicative group $\mathbb{Z}_p^*$, addition of elements in this group is well defined everywhere as well—but modulo $p-1$ of course. So, given a prime p and a generator g, we still have $\mu_g(u+v) = g^{u+v} \bmod p$ for any two integers u and v. The inverse ψ of the power map μ is of course the discrete log function, i.e., if we set $j = g^i$, then $j = g^{\psi(j)}$ which remains well defined in $\mathbb{Z}_p^*$ with the caveat that $\psi(1) = p-1$ and thus $\psi(ij) = \psi(i) + \psi(j) \bmod (p-1)$. This consideration makes maximal use of the fact that $(\mathbb{Z}/p\mathbb{Z}, +, \cdot)$ is a ring, which we will exploit in the following.

One immediate question to ask is if there exists a generator g such that the map μ_g has a fixpoint. This question is normally attributed to Brezolis (see Guy [2, Sect. F9]), and was solved in the affirmative by Levin et al. [4]. I first became aware of such a fixpoint when I was doing Peirce's "Card Trick" (see [6]) which I had read about in Martin Gardner's corner in Scientific American July 1978, which illustrates that $\mathbb{Z}_{13}^*$ is cyclic and generated by 2, where I observed that $2^{10} = 10$.

So at some much later stage I began thinking more seriously about fixpoints and the orbits in general and only through the feedback of the reviewer of an earlier version learned that quite extensive work has been carried out on this theme within the last decade. The original aim of the present note was to discuss to what extent this simple construction has any impact on or consequences for the solution to the Discrete Log Problem (DLP). But during the analysis we obtained a complete classification of all fixpoints of the map μ_g when g runs through the set of generators, as well as the exact number of generators g for which a particular element i is a fixpoint of μ_g.

For a general discussion of the DLP, we refer to [5, Sect. 3.6]. For other significant but independent recent contributions, see [1] and [3].

Recall that Fermat's Little Theorem states that $\mathbb{Z}_p^*$ is cyclic, and the number of generators is $\phi(p-1)$, where ϕ is Euler's function. This function is multiplicative on relatively prime factors of $p-1$, and $\phi(q^r) = (q-1)q^{r-1}$ for q a prime.

We note that there is no general formula for generators, and it is not even known if there are infinitely many primes for which 2 is a generator. However, given a specific prime p it is easy to find a generator if you know the factorization of $p-1$ using trial and error, and once you know one generator g, you can determine them all as the set $\{g^s \mid \gcd(s, p-1) = 1\}$. In the following, given p and g, we denote a specific power map defined by g with μ_g.

2 Basic properties

So with the notation above, μ is an element of the symmetric group on $p-1$ letters, Σ_{p-1}. Any permutation in the symmetric group can be written in a unique way as a product of disjoint cycles, each cycle corresponding to an orbit of ϕ. A cycle is written as $(a_1, a_2, a_3, \ldots, a_n)$, the interpretation of which is that a_i is mapped to a_{i+1} for $i < n$ while a_n is mapped to a_1. If the permutation has several orbits, we just concatenate the cycles and continue to read from left to right say (not that it matters as disjoint cycles commute). With this notation, given any orbit of length n and an element a_i in the orbit, the previous element in the orbit a_{i-1} is simply the discrete log of its successor with respect to the base g. But moving forward one step in the orbit from s is calculating the image g^s, so starting with $t = g^s$, applying the power map n times yields the discrete log s of t.

We thus have the following: Given g and an exponent s, s and g^s lie in the same orbit, and the length of the orbit in which s lies under the power map μ_g is the (maximal) number

of iterations of μ_g needed to find the discrete log of g^s. We say 'maximal' as it is possible to reduce the complexity of this exercise, e.g., by pre-calculating g to all powers of 2 less than $p-1$. More on that later.

3 A small example and some fundamental observations

Let us begin with a very simple example to illustrate this approach, inspired by Peirce's Card Trick. Let $p = 13$, so $\mathbb{Z}_p^*$ has order 12. By trial and error, we see that 2 is a generator. Table 1 describes the map $\mu_2 : i \mapsto 2^i \bmod 13$, which is an element of Σ_{12}. As $\phi(12) = 4$, we have four generators, namely g^a where $a \in \{1, 5, 7, 11\}$. Writing this permutation as a product of disjoint cycles, we get $\mu_2 = (1, 2, 4, 3, 8, 9, 5, 6, 12)\ (7, 11)(10)$. It obviously does not depend on the orders of the elements, multiplicative or additive, whether they lie in the same orbit or not, in that elements of all possible orders, multiplicative as well as additive, occur in the long orbit of μ_2. This is one point we shall revert to. In any event, we see that this map has three orbits, one of order 9, one of order 2, and a fixpoint orbit, and thus has order 18, the lcm of all individual orbit lengths.

Let us consider the remaining three maps (see Table 2). Before we do so, we point out that all images under the four power maps of one particular element, i.e., all elements in a column, have the same multiplicative order, namely the additive order of the chosen element.

Another elementary observation is that once you have computed one power map, e.g., μ_2, you can read all the other power maps directly off the compilation of this map. Thus $\mu_6(i) = \mu_2(5i)$ as $6 = 2^5$ (not to be confused with the entirely different map μ_6^5). The map defined by 6 is $(1, 6, 12)(2, 10, 4, 9, 5)(3, 8)(7)(11)$ with 3 orbits of length larger than 1 and 2 fixpoints. The map defined by 11 is $(1, 11, 6, 12)(2, 4, 3, 5, 7)(8, 9)(10)$ with 3 orbits of length larger than 1 and 1 fixpoint. Finally, the map defined by 7 is $(1, 7, 6, 12)(2, 10, 4, 9, 8, 3, 5, 11)$ with 2 orbits of order 4 and 8, respectively, and no fixpoints. Properties of the four permutations of $\mathbb{Z}_{13}^*$ are summarized in Table 3. Other characteristics to consider for each element s are the maximal length of an orbit for some power map containing s as well as the minimal length; see Table 4.

Table 1 The power map μ_2 on $\mathbb{Z}_{13}^*$

Exponent	1	2	3	4	5	6	7	8	9	10	11	12
Power of 2	2	4	8	3	6	12	11	9	5	10	7	1
Mult. order	12	6	4	3	12	2	12	3	4	6	12	1
Add. order	6	3	3	4	2	1	12	4	12	6	12	12

Table 2 The power maps $\mu_2, \mu_6, \mu_7, \mu_{11}$ on $\mathbb{Z}_{13}^*$

Exponent	1	2	3	4	5	6	7	8	9	10	11	12
Additive order	12	6	4	3	12	2	12	3	4	6	12	1
μ_2	2	4	8	3	6	12	11	9	5	10	7	1
μ_6	6	10	8	9	2	12	7	3	5	4	11	1
μ_{11}	11	4	5	3	7	12	2	9	8	10	6	1
μ_7	7	10	5	9	11	12	6	3	8	4	2	1

Table 3 Properties of μ_2, μ_6, μ_7, μ_{11} on $\mathbb{Z}_{13}^*$

Generator	Type in Σ_{12}	Order	#Fixpoints
2	$1^1 2^1 9^1$	18	1
6	$1^2 2^1 3^1 5^1$	30	2
11	$1^1 2^1 4^1 5^1$	20	1
7	$4^1 8^1$	8	0

Table 4 Maximum and minimum lengths of orbits containing an element of $\mathbb{Z}_{13}^*$

Element	1	2	3	4	5	6	7	8	9	10	11	12
Maximum	9	9	9	9	9	9	5	9	9	8	8	9
Minimum	3	5	2	5	5	3	1	2	2	1	1	3

Table 5 Properties of the power maps on $\mathbb{Z}_{23}^*$

Generator	Type in Σ_{22}	Order	#Fixpoints	#elements in orbits of length $\leq 3, 4$
5	$1^2 3^1 4^1 13^1$	156	2	(5,9)
10	$1^2 2^1 8^1 10^1$	40	2	(4,4)
20	22^1	22	0	(0,0)
17	$1^1 2^1 3^1 4^1 12^1$	12	1	(6,10)
11	$1^5 2^1 3^1 6^2$	6	5	(10,10)
21	$4^1 5^1 13^1$	260	0	(0,4)
19	$2^1 4^1 16^1$	16	0	(2,6)
15	$1^2 3^1 5^1 12^1$	60	2	(5,5)
7	$1^1 7^1 14^1$	14	1	(1,1)
14	$7^2 8^1$	56	0	(0,0)
Total			13	(33,49)
Average			1.3	(3.3,4.9)

So, for $p = 13$, choose 6 as a generator, and suppose $j = 8$. We need to find i where $6^i = j$. We consider the orbit containing 8: 8 is mapped to 6^8 mod 13 which happens to be 3, and 3 is mapped to 6^3 mod 13—which is 8. So we have solved the problem just using one exponentiation because the j that was chosen for the base 6 happened to lie in a short orbit of length 2.

Had we chosen the base 2, the length of the orbit would be much larger, namely 9. Had we chosen 11, it would be short again, of length 2, and finally, had we chosen 7, it would be of length 8.

Another general question is to ask: Does it ever occur that the map μ_g is transitive? In fact it does, but apparently relatively rarely. Let us consider one more example for which this in fact happens, namely 23; see Table 5. In particular, we observe that there is exactly one permutation which is transitive on $\mathbb{Z}_{23}^*$. There are altogether 13 fixpoints and 4 orbits of length 2.

Let us summarize the most important initial observations:

Lemma 1 *Let p be an odd prime and let g be a fixed generator of $\mathbb{Z}_p^*$. Let h be any generator and let $h = g^s$. Then $\mu_h(i) = \mu_g(si)$, and for any i the order of $\mu_h(i)$ in $\mathbb{Z}_p^*$ is the order of i in $(\mathbb{Z}/(p-1)\mathbb{Z}, +)$.*

However, and this is the most important fact, different power maps on $\mathbb{Z}_p^*$ have different number of orbits, and many even have fixpoints as mentioned, whereas transitive power maps appear to be relatively rare. In addition, different elements in one orbit typically have completely uncorrelated orders, with a few exceptions, the obvious one being that the orbit containing 1 always maps $(p-1)/2$ to $p-1$ and $p-1$ to 1 (see Theorem 10 for more).

The examples above, however small, suggest that perhaps some choices of generators and exponents are better than others as far as solving the DLP is concerned. Indeed, for the prime 13, 2 and 7 are better choices as generators than are 6 and 11. They have longer orbits, and even if you choose 2 as a generator and $s = 8$, then the next element in the orbit is $3 = 2^8$, and you then find s in one calculation as $8 = 2^3$, as 3 and 8 lies in an orbit of length 2. Even better (or worse) is the choice of $s = 10$, as this is a fixpoint for μ_2 as we have seen. At the same time, we notice that $s = 3$ is not an easy case if the generators 2 or 7 are chosen instead, as the orbit lengths are 9 and 8 respectively.

Likewise for the prime 23, there is a significant difference between the generator 20 and the generator 11.

4 Questions

The discussion above only serves as illustration of what we are after. The real question is, how does this work for large primes? Obviously, if the occurrence of short orbits generalizes, the DLP is not always as hard to solve as hitherto anticipated, and if the DLP is to be used in public-key cryptography, certain generators and certain keys could be much weaker than others and should be avoided either by construction or because one can prove that with a good random bit generator, short orbits are infeasible.

So as far as the use in cryptography is concerned, an important question for general primes is: Can we either give sufficient conditions to avoid generators with too many orbits of small length, say of the order of magnitude of $o(\log^k p)$ for small k, or if not can we prove that if the generator g and the secret key s are chosen sufficiently randomly, the probability that the order of the orbit containing g^s is small is negligible? We shall not be able to answer that here.

Now, rephrasing an earlier observation, given $j = g^i \bmod p$ for a generator g, one approach to finding i would be to replace the generator g by another generator to reduce the complexity of the DLP. So assume we know a generator $h = g^a$ with all or a substantial part of all orbits sufficiently short. The first step is to find b satisfying $g = h^b \bmod p$. This is done by means of the orbit of h containing g, that is $(g, h^g, \ldots)$, the last element of which will be b. Thus $j = h^{bi} \bmod p$, and as we know b, finding i is equivalent to finding bi. But we know bi lies in the orbit containing h^{bi}, so again we calculate all the elements in this orbit and again the element in the orbit just before h^{bi} is bi, and we have i.

Thus one key question—taking this approach—is: Given j as above, can we find a generator g^a such that j lies in a sufficiently short orbit with respect to g^a? Or stated in a more controversial manner: Are there any trapdoor approaches with clever choices of generators?

5 A characterization of fixpoints

We need a few more tools. Define A as the set of elements in $\mathbb{Z}_p^*$ that are relatively prime to $p-1$. We now define a composition $\circ$ on this set, where $a \circ b$ is defined as $ab \bmod p - 1$. Then A is a group under this composition, and isomorphic to the automorphism group $\mathrm{Aut}(\mathbb{Z}_p^*)$

of $\mathbb{Z}_p^*$. If the prime divisors of $p-1$ are $q_1 = 2, q_2, \ldots q_r$, then $\mathbb{Z}_p^*$ is the direct product of its corresponding Sylow q_j-subgroups S_1 through S_r, and likewise $\mathrm{Aut}(\mathbb{Z}_p^*)$ is the direct product of the automorphism groups of S_1 through S_r. Moreover, if S is a cyclic q-group of order q^k for some prime q, then the automorphism group is cyclic of order $q^{k-1}(q-1)$ for q odd, and for $q = 2$ and $k > 2$ it is the direct product of a group of order 2 and a cyclic group of order 2^{k-2}.

Before we start proving our first results, we need the following elementary result, which may not be known to all readers.

Lemma 2 *Let G be a finite cyclic group, and let H be a subgroup. Then any automorphism of H may be extended to an automorphism of G.*

Proof Consider first a p-group G for some prime p, and write G multiplicatively. Let g be a generator of G, and let $h = g^d$ be a generator of H, where d may be chosen to equal the order of G/H. Any automorphism of H is uniquely determined by what it maps h too, say h^a, where obviously a must be prime to p, since h^a must be a generator of H too. But this naturally extends to an automorphism of G namely the one defined by mapping g to g^a.

In the general case, G is the direct product of its various Sylow p-subgroups $G_1, \ldots, G_k$, and so is H of its various Sylow p-subgroups $H_1, \ldots, H_k$ (where some may be of order 1) and $H_i \leq G_i$. As obviously $\mathrm{Aut}(G) = \mathrm{Aut}(G_1) \times \cdots \times \mathrm{Aut}(G_k)$, the general result follows immediately from the special result. □

Theorem 3 *Let p be a prime and $0 < i < p$ and consider i as an element in the multiplicative group $\mathbb{Z}_p^*$. Then the following are equivalent:*

(i) *$ord(i) = (p-1)/\gcd(i, p-1)$.*
(ii) *There exists a generator g of $\mathbb{Z}_p^*$ such that $g^i = i$, i.e., i is a fixpoint for the permutation μ_g.*
(iii) *g^i and i have the same order for some generator g.*

Proof In general, if $h^j = i$ for some $i \in \mathbb{Z}_p^*$ and generator h, then the order of i is $(p-1)/\gcd(j, p-1)$. In particular, (i) immediately follows from (ii).

Conversely, assume (i), and let $d = \gcd(i, p-1)$. By assumption, i lies in the one and only subgroup H of $\mathbb{Z}_p^*$ of order $(p-1)/d$ and in fact is a generator of this subgroup. This subgroup is generated by g^d as well, so there exists j such that $g^{dj} = i$. As i has order $(p-1)/d$ by assumption, j is prime to $(p-1)/d$. Thus $g^d \mapsto g^{dj}$ is an automorphism of H, which by Lemma 2 may be extended to an automorphism $\psi : g \mapsto g^a$, i.e., $g^{dj} = g^{da}$ (which is just another way of stating that j can in fact be chosen to be prime to $p-1$ rather than just prime to $(p-1)/d$). Replacing the generator g with the generator $\psi^{-1}(g)$, we have found a generator g such that $g^d = i$. Now let $i = ed$. As e is prime to $p-1$, there exists an f such that $ef \equiv 1 \pmod{p-1}$, and in particular $g^{ef} = g$. Hence $g^{efd} = (g^f)^i = i$, and as g^f is a generator too, we have proved (ii).

Now (iii) is a trivial consequence of (ii). Conversely, as the order of g^i always is $(p-1)/\gcd(i, p-1)$ as remarked above, (i) follows from (iii). □

As a consequence, given g^i, find the order of g^i, which is easy if the factorization of $p-1$ is known. If the order of g^i is $p-1$, we know that (i) i is prime to $p-1$; and (ii) there exists an s prime to $p-1$ such that $h = g^s$ and $h^i = i$. But we can go further even if we do not know the factors of $p-1$.

Denote the additive order and the multiplicative order of an element i by $\mathrm{ao}(i)$ and $\mathrm{mo}(i)$ respectively. Obviously, the set of i's such that $\mathrm{mo}(i) = \mathrm{ao}(i)$ equals the set of g^j's such that $\mathrm{mo}(g^j) = \mathrm{ao}(g^j)$ for g a generator.

Definition 1 Any i satisfying the property that there exists a generator g such that $g^i = i$ is called a *discrete log fixpoint*, or just *fixpoint* of $\mathbb{Z}_p^*$. Theorem 3 characterizes all the discrete log fixpoints effectively if we know all the prime factors of $p-1$ as the set of elements i for which $\mathrm{ao}(i) = mo(i)$. We call this the *common order* of i. For i a fixpoint, we define the *multiplicity* $\mathrm{mult}(i)$ of i as the number of generators g such that $g^i = i$.

We can now reformulate Theorem 3.

Theorem 4 *The following are equivalent:*

(1) *i is a fixpoint.*
(2) *$mo(i) = ao(i)$.*
(3) *There exists a generator g such that i and g^i have the same multiplicative order.*
(4) *There exists a generator h and a j such that $i := h^j$ and j have the same additive order.*

Proof Theorem 3 states that (1), which is (i) above, is equivalent to (2), which is (ii) above. Assume (2), then g in (3) is the g for which $i = g^i$. Now assume (3). As always $\mathrm{mo}(g^i) = \mathrm{ao}(i)$, it follows that $\mathrm{mo}(i) = \mathrm{ao}(i)$ as well, so (2) and (3) are equivalent. Moreover, (1) implies (4) with h a generator for which i is a fixpoint. Finally, assume (4). Then it follows that $\mathrm{ao}(h^j) = \mathrm{ao}(j) = \mathrm{mo}(h^j)$, so $i := h^j$ is a fixpoint as (2) implies (3). □

Theorem 5 *Consider the group $\mathbb{Z}_p^*$. Let i be a fixpoint of common order $(p-1)/q$ for some q, and let $q_1, q_2, \ldots, q_k$ be the different prime divisors with multiplicities as divisors of q equal to $r_1, r_2, \ldots, r_k$ respectively, where the first t primes divide $(p-1)/q$, and the rest are relatively prime to $(p-1)/q$. Then*

$$mult(i) = \prod_{j \le t} q_j^{r_j} \cdot \prod_{j > t} q_j^{r_j - 1}(q_j - 1).$$

Proof Let $h^i = g^i = i$ for some generators g and h. It follows that gh^{-1} lies in the kernel I of the homomorphism ψ of $G = \mathbb{Z}_p^*$ into itself defined by $\psi(x) = x^i$. This kernel is the unique subgroup of order q, and $I = H \times L$, where the prime divisors of H are the first t q_i's above, and those of L are the remaining q_j's. Furthermore, by choice a Sylow q_i-subgroup of H is a proper subgroup of a Sylow q_i-subgroup K_i of G, whereas each Sylow q_j-subgroup of L is a Sylow q_j-subgroup L_i of G. Thus $L = L_{t+1} \times \cdots \times L_k$ and if we set $K = K_1 \times \cdots \times K_t$, then $G = K \times L$, $H \le K$, and the order of K/H is divisible by all the first t primes q_i. Let the generator g equal $g_1 g_2$ where g_1 is a generator of K and g_2 is a generator of L.

We now claim that if x lies in H, then $g_1 x$ is a generator of K as well. Writing g_1 and x respectively as the unique product of elements in the Sylow subgroups of H we see that it suffices to prove this within any Sylow q-subgroup S of K, i.e., any of the groups K_1 through K_t. Let this have order q^s, and let $S \cap H$ have order q^r, where $r < s$. But then if a is the component of g_1 in S, it has order q^s, so for any y in $S \cap H$, $(ax)^{q^{s-1}} = a^{q^{s-1}}$—which is not 1. Thus ay is a generator of S, and the claim follows.

Next we need to determine the $x \in L$ for which $g_2 x$ is a generator of L. Again we need only consider what happens with the component b of g_2 and z of x in a Sylow q-subgroup for some prime q dividing the order of L. let T be of order q^r and let U be the unique subgroup of index q. Thus T is generated by b, and bz is a generator iff bz does not lie in U. But bz lies in U iff $bU = z^{-1}U$, which in particular implies that $z \notin U$. So the number of elements in T for which bz is a generator is $q^{r-1}(q-1)$, and the claim follows. □

Example 1 Let $p = 43$. Then 7 is a fixpoint of common order 6. This is in fact easy to see, once you have verified the order of 7 is 6. Indeed, $\mathbb{Z}_p^* = S \times T$, where S is of order 6 generated by 7, and T is of order 7. Now as 7 has order 6, $7^7 = 7$, and for every element $t \in T$, $t^7 = 1$. A generator of T is 4, so the 6 non-trivial elements of T are $\{4, 16, 21, 41, 35, 11\}$, so the generators of $\mathbb{Z}_p^*$ for which 7 is a fixpoint are exactly the elements in the coset $7 \cdot T$ different from 7, i.e., $\{28, 26, 18, 29, 30, 34\}$.

Corollary 6 *In particular, the following holds:*

(1) *Let i be a fixpoint of common order $p-1$. Then the multiplicity of i is* 1.
(2) *Assume $p \equiv 1 \pmod 4$, and let i be a fixpoint. Assume i is an even number. Then the multiplicity of i is even.*
(3) *Assume $p \equiv 3 \pmod 4$ and let i be a fixpoint of common order $(p-1)/2$. Then the multiplicity of i is* 1.
(4) *Assume $p - 1 = 2q$, where q is an odd prime. Let i be different from $(p-1)/2$ and $(p+1)/2$. Then i is a fixpoint iff $p - i$ is. Furthermore, $(p+1)/2$ may be a fixpoint (but $(p-1)/2$ of multiplicative order 2 obviously not).*

Proof (1) If i is of common order $p-1$, the map $g \mapsto g^i$ is an automorphism.
(2) This follows from Lemma 7, as $g^i = (p-g)^i = (-g)^i$ for all generators g.
(3) Let $g^i = i = h^i$ for generators g, h. Then gh^{-1} lies in the kernel $K = \langle \gamma \rangle$ of the endomorphism $\phi(a) = a^i$ for $a \in \mathbb{Z}_p^*$, which has order 2 if i has common order $(p-1)/2$. So either $g = h$ or $g = h\gamma$, where γ is the element of order 2. But in the latter case, as $(p-1)/2$ is odd, $g^i = h^i\gamma$, a contradiction.
(4) Assume i is a fixpoint. By assumption and choice, the common order of i is either q or $2q$, and by assumption as well $p-i$ is neither $p-1$ nor $(p-1)/2$, so $\mathrm{mo}(i)$ is either q or $2q$, and likewise with $\mathrm{ao}(i)$. Assume $\mathrm{mo}(i) = 2q$. Then $\mathrm{ao}(p-i) = q$ by Lemma 7. Moreover, i is odd, so $p-i$ is even, and it follows that $\mathrm{ao}(p-i) = q$ as well. The case where the common order of i is q is handled in a similar fashion. Assume next that $p-i$ is a fixpoint. Then so is $p-(p-i) = i$ by what we have just proved. □

6 On generators

Lemma 7 *Let p be a prime*

(i) *If $p \equiv 1 \pmod 4$ then g is a generator of $\mathbb{Z}_p^*$ iff $p-g$ is as well.*
(ii) *If $p \equiv 3 \pmod 4$ then g is a generator of $\mathbb{Z}_p^*$ iff $g^{(p+3)/2}$ is as well.*
(ii) *If $p \equiv 3 \pmod 4$ then g has multiplicative order $p-1$ iff $p-g$ has multiplicative order $(p-1)/2$.*

Proof The element $p-g$ is obviously equal to $-g \bmod p$, or g multiplied by the involution in $\mathbb{Z}_p^*$. If $p \equiv 1 \pmod 4$, then 4 divides the order of g, and it follows that g and $-g$ have the same order. If $p \equiv 3 \pmod 4$, and g is a generator, then g^2 has order $(p-1)/2$ which is odd, and $g^{(p-1)/2}$ has order 2, and thus $g^{2+(p-1)/2} = g^{(p+3)/2}$ has order $2 \cdot ((p-1)/2) = p-1$. Conversely, if g is not a generator then no power of g is either. □

Lemma 8 *Let g be a generator of $\mathbb{Z}_p^*$.*

(a) *Assume $p \equiv 3 \pmod 4$. Then $g \mapsto g^{(p+3)/2}$ defines an automorphism Ψ of $\mathbb{Z}_p^*$, and*

$$g,\ g^{(p+3)/2},\ g^{(p+7)/2},\ g^{(p+15)/2},\ \ldots,\ g^{(p+2^{k-1}-1)/2} \tag{1}$$

are all generators of $\mathbb{Z}_p^*$ *with* $g = g^{(p+2^k-1)/2}$, *where* k *is the order of the automorphism* Ψ *and in particular divides* $\phi(p-1)$. *The inverse of* Ψ *is the map* $g \mapsto g^{(p-3)/4}$ *or* $g^{(p-1)/4}$ *respectively, where in the first case* $p \equiv 7 \pmod 8$ *and in the second case* $p \equiv 3 \pmod 8$.

(b) *Assume* $p \equiv 1 \pmod 4$, *and let* $p - 1 = 2^k r$, *where* r *is odd. Choose the smallest multiple* s *of* r *which is* $1 \bmod 2^k$. *Then* $\phi_i(g)$ *defined by* $\phi_i(g) = g^{(s+2^i) \bmod (p-1)}$ *is a non-trivial automorphism for all* $i > 0$. *Notice that* i *is not necessarily bounded by* k.

Proof (a) That all the elements listed in (1) are generators follows by repeated application of (2) above as for any h in the list $h^{(p+3)/2}$ is the next in the list. Indeed, the exponent in $(g^{(p+2^i-1)/2})^{(p+3)/2}$ is $(p + 2^i - 1)(p + 3)/4 = p(p + 2^i + 1)/4 + (p + 3(2^i - 1))/4$ and thus

$$\left(g^{(p+2^i-1)/2}\right)^{(p+3)/2} = g^{(p+2^i+1+p+3(2^i-1))/4} = g^{(2p+2^{i+2}-2)/4} = g^{(p+2^{i+1}-1)/2}.$$

(b) Let $g = ab$ where a is a generator of the Sylow 2-subgroup and b a generator of the complement of order r. Then $g^s = a$, $b^s = 1$, and thus $\phi_i(g) = g^{s+2^i} = a^{1+2^j} b^{2^j}$ for $i > 0$ and thus is the composition of an automorphism of the Sylow 2-subgroup and of its complement. Obviously, $s + 2^j$ can be replaced by $(s + 2^j) \bmod (p - 1)$. □

Example 2 Let $p = 31$. $\mathrm{Aut}(\mathbb{Z}_{31}^*)$ is isomorphic to $\mathbb{Z}_3^* \times \mathbb{Z}_5^*$, and starting with a generator g, Ψ above yields half of the generators g^{17}, g^{19} and g^{23} (where the next is $g^{31} = g$). Starting with any other generator not in this list, say g^7, we obtain the remaining four, which are the inverses of those above.

For $p = 43$, $\mathrm{Aut}(\mathbb{Z}_{43}^*)$ is isomorphic to $\mathbb{Z}_3^* \times \mathbb{Z}_7^*$, and the sequence we get is g, g^{23}, g^{25}, g^{29}, g^{37}, g^{11}, and again we have half of the generators.

For $p = 19$, $\mathrm{Aut}(\mathbb{Z}_{19}^*)$ is isomorphic to $\mathbb{Z}_9^*$, the multiplicative subgroup of $\mathbb{Z}/9\mathbb{Z}$ and thus cyclic, and the sequence is the entire list of generators, g, g^{11}, g^{13}, g^{17}, g^7, g^5.

Likewise for $p = 23$. Then we get g, g^{13}, g^{15}, g^{19}, g^5, g^{21}, g^9, g^7, g^3, g^{17}, all of the 10 generators.

For $p = 127$, the automorphism group is the direct product of two cyclic groups each of order 6, and thus of exponent 6, and accordingly the sequence we get is of length 6: g, g^{65}, g^{67}, g^{71}, g^{79}, g^{95}.

Note that another way of describing the sequence of generators above is g, g^{a+2}, g^{a+4}, g^{a+8}, …, where g^a is a generator of the Sylow 2-subgroup.

Example 3 Let $p = 61$. So $p - 1 = 15 \times 4$ and $s = 45$. The sequence of generators we get is g, g^{47}, g^{49}, g^{53}.

Let $p = 101$. So $p - 1 = 25 \times 4$, and here we get g, g^{27}, g^{29}, g^{33}, g^{41}, g^{57}, g^{89}, g^{53}, g^{81}, g^{37}, g^{49}, g^{73}, g^{21}, g^{17}, g^9, g^{93}, g^{61}, g^{97}, g^{69}, g^{13}, half of all the generators.

7 Orbits of length 2

So let g be a generator and assume $g^a = b$ and $g^b = a$. It follows that $g^{ab} = b^b = a^a$. In particular, $\mathrm{mo}(b) = (p-1)/\gcd(a, p-1)$, and $\mathrm{mo}(a) = (p-1)/\gcd(b, p-1)$, but a and b do not necessarily have to have the same order.

Unfortunately it is not true that the condition $b^b = a^a$ for two elements in $\mathbb{Z}_p^*$ is sufficient to ensure that there exists a generator g such that $g^a = b$ and $g^b = a$.

Another condition to consider is that if $\{a, b\}$ is an orbit for g, then $g^{a+b} = ab$. So one might ask conversely, if $g^{a+b} = ab$ for a generator g, does it follow that $\{a, b\}$ is an orbit or that a and b are both fixpoints?

In Theorem 10 we shall prove one general albeit special occurrence of orbits of length 2 whenever 2 is a generator.

8 The orbit containing 1

Obviously, this contains $p - 1$ as well, as this is mapped to 1.
Which element is mapped to $p - 1$? This has to be $(p - 1)/2$, obviously. Next we observe that if 2 is a generator, then $(p - 3)/2$ is mapped to $(p - 1)/2$. Obviously 2 is a generator of $\mathbb{Z}_p^*$ iff $(p + 1)/2$ is as they are each other's inverse.

Lemma 9 *Assume* 2 *is a generator, and let* g *be the chosen generator. Let* $2 = g^s$ *and let* t *be the inverse of* s *modulo* $p - 1$. *Then* $(p - 1)/2 - s$ *modulo* $p - 1$ *is mapped to* $(p - 1)/2$ *under* μ_g.

Proof We are looking for j such that $g^j = (p - 1)/2$, or equivalently $g^{j+s} = p - 1$. This implies that $j + s = (p - 1)/2$ modulo $p - 1$. □

Theorem 10 (a) *Assume* 2 *is a generator.*

(i) *Let* $g = 2^{p-2} = (p + 1)/2$ *be the inverse of* 2. *Then the orbit containing* 1 *consists of the four elements* $((p + 1)/2, (p - 1)/2, p - 1, 1)$.
(ii) *Let* $p \equiv 1 \pmod 4$ *and* $g = 2^{(p+1)/2}$. *Then* $g = p - 2$ *and the orbit containing* 1 *consists of the four elements* $(p - 2, (p - 1)/2, p - 1, 1)$.
(iii) *Let* $p \equiv 1 \pmod 4$ *and* $g = 2^{(p-3)/2}$. *Then* $g = (p - 1)/2$ *and the orbit containing* 1 *consists of the three elements* $((p - 1)/2, p - 1, 1)$.
(iv) *Let* $p \equiv 3 \pmod 4$ *and* $g = 2^{(p-5)/2}$. *Then* $g = (p - 1)/2$ *and the orbit containing* 1 *consists of the four elements* $((3p - 1)/4, (p - 1)/2, p - 1, 1)$.
(v) *Let* $p \equiv 3 \pmod 4$ *and* $g = 2^{(p+3)/2}$. *Then* $g = p - 4$ *and the orbit containing* 1 *contains the elements* $((p - 3)/4, (p - 1)/2, p - 1, 1, p - 4)$.
(vi) *Let* $g = 2$. *Then* $((p + 1)/2, p - 2)$ *is an orbit of length 2.*

(b) *Assume* $(p - 1)/2$ *is a generator. Then* $((p - 1)/2, p - 1, 1)$ *is an orbit for the corresponding power map.*

Proof (a) (i) As $(p + 1)/2$ is the inverse of 2 modulo p, the image of 1 with that choice of the generator is g.
(ii) As $2^{(p-1)/2} = p-1$, $g = 2^{(p+1)/2} = p-2$. Hence $g^{p-2} = -2^{p-2} = (p-1)/2$ by (i).
(iii) By (ii), $4g = p - 2$ with this choice of g, and it follows that $g = (p - 1)/2$.
(iv) With this choice of g, $4g = p - 1$, so $2g = (p - 1)/2 = -(p + 1)/2$, and thus $g = -(p + 1)/4 = (3p - 1)/4$. Furthermore, $g^{(3p-1)/4} = -2^{p-2} = (p - 1)/2$.
(v) Here $g = 2^{(p+3)/2} = -2^2$. Now let $(p - 4)^i = (p - 1)/2$. Assume i is even. Then $(p-4)^i i = 2^{2i} = (p-1)/2$, or $2^{2i+1} = p-1$ which implies that $2i+1 = (p-1)/2$ modulo $p - 1$, or $2i = (p-3)/2$. Hence $i = (p-3)/4$. Conversely, we see this will do as a solution to the equation we aimed to solve, and as there is only one solution, this must be the one!
(vi) As we saw in the proof of (ii), $2^{(p+1)/2} = p - 2$ (regardless of the choice of p). Likewise, by (i), $2^{p-2} = 2^{-1} = (p + 1)/2$.
(b) The power map defined by $(p - 1)/2$ maps 1 to $(p - 1)/2$, which obviously is mapped to the element of order 2. □

Example 4 We saw that (7,11) is an orbit for the power map μ_2 for the prime 13. So this was perfectly predictable.

9 Restrictions to subgroups of $\mathbb{Z}_p^*$

Before we revert to $\mathbb{Z}_p^*$, let G be a finite cyclic group generated by g, and let H be a subgroup of order k and index n, the number of cosets of H in G, and assume that k and n are relatively prime.

Then g^n is a generator of H, and $H = \{g^n, g^{2n}, \ldots, g^{kn} = 1\}$, and the cosets of H are gH, $g^2H, \ldots, g^{n-1}H, H$. For another choice of cosets, we can choose $g^kH, g^{2k}H, \ldots, g^{(n-1)k}H$, and if K is generated by g^k, then $G = H \times K$.

Let p be a prime, and q a divisor of $p-1$ such that q and the index $r = (p-1)/q$ are relatively prime. Let Q be the subgroup of order q. One question is if we can define power maps in a similar fashion on the subgroup H rather than G.

Consider first the case where $r = 2$. Here $h = g^2$ is a generator of H, so the power map defined by h is a map of G onto H, and each element in H is the image of exactly two elements as $g^a = g^{a+(p-1)}$. However, this does not necessarily imply that the restriction of this map to H is injective, even in the case where H is the subgroup of index 2 in $\mathbb{Z}_p^*$, for any prime congruent to 3 mod 4. However, as we are in charge we can define a modified power map μ^* as follows. We start with a generator g of G. Then g^2 is a generator of H. Thus $H = \{g^{2i} | i = 1, 2, \ldots, q\}$, so we define $\mu^*(i) = g^{2i}$ for $i \leq q$ whereas for $i > q$ we define $\mu^*(i) = \mu(p-i)$.

Lemma 11 *Let p be a prime, which is 3 mod 4, and let g be a generator of $\mathbb{Z}_p^*$. Let H be the subgroup of index 2 generated by g^2. Then the power map μ^* defined above is a permutation of H.*

Proof Obviously μ^* maps onto H, so all we need to prove is that its restriction to H is injective. To this end, we observe that H is the disjoint union of the set A of elements less than or equal to q and the complement B, and μ^* is injective on A as $g^{2i} \neq g^{2j}$ for $i \neq j$ and $i, j \leq q$. If $i \neq j$ and $i, j > q$, $p - i \neq p - j$ are both bounded by q, so again the images under μ^* are different. Finally let $i \leq q < j$ and assume $\mu^*(i) = \mu^*(p-j)$. Then $i = p - j$ as they are both bounded by q, so i and $-j$ are congruent modulo p. But i and j are both elements in H, so $-j$ is an element of $-H$ as well, a contradiction. □

Example 5 To demonstrate that this is just as useful, consider again $p = 47$, and let H be the subgroup of order 23. It turns out that 40 is a generator for the whole group, and thus $40^2 = 2$ (funnily enough) is a generator of H. It is convenient to think of the one coset different from H as $-H$, as $-1 = p - 1$ is not in H.

Consider the map μ defined as powers of 2, which yields the elements of H. Now extend this to a map μ^* of G onto H by defining $\mu^*(i) = \mu(47 - i)$ for $i > 23$.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
μ	2	4	8	16	32	17	34	21	42	37	27	7	14	28	9	18	36	25	3	6	12	24	1
μ^*					15		13		5	10	20			19			11	22				23	

So the second row identifies the elements in H. The orbit containing 1 starts with $(1, 2, 4, 16, 18, 25, \ldots)$. As $25 > 23$, 25 is mapped to $\mu(22) = 24$, and 24 is mapped

to $\mu(23) = 1$. So the first orbit is (1,2,4,16,18,25,24). The first element in H not in this orbit is 3, and the orbit of 3 starts with $(3, 8, 21, 12, 7, 34, \ldots)$. The image of 34 is the same as the image of 13 which is 14 and so on, so the orbit is (3,8,21,12,7,34,14,28). The remaining orbits are (6,17,36,20), (9,42,32) and (37), i.e., μ^* is of type $1^1 3^1 4^1 7^1 8^1$.

Reverting to the notation above, we get that under the power map defined by g^n, all elements in hK for $h \in H$ maps to g^{nh} and thus the power map again induces a permutation on H.

The next question is, does this generalize in some way rather to the case where the index r of H in G is arbitrary? We have looked very carefully—and optimistically, too—into this, but without success.

Personal note

The last time I had a chance to talk mathematics with Scott, I showed him an earlier version of the present work. He got quite excited about this new approach to the DLP in $\mathbb{Z}_p^*$ as it reveals that there are non-obvious weak keys for various cryptographic schemes over this group. Scott encouraged me to make this public, but I had not anticipated that when I finally got around to it, it would be caused by his premature death. Many of us miss him as a good friend and an excellent mathematician and cryptographer.

Acknowledgments I am grateful to the reviewer for making me aware of some relevant publications and their results.

References

1. Bourgain J., Konyagin S., Shparlinski I.: Product sets of rationals, multiplicative translates of subgroups in residue rings and fixed points of the discrete logarithm. In: International Mathematics Research Notices (2008).
2. Guy R.: Unsolved Problems in Number Theory. Problem Books in Mathematics, 3rd edn. Springer, New York (2004).
3. Holden J., Moree P.: Some heuristics and results for small cycles of the discrete logarithm. Math. Comput. **75**, 419–449 (2006).
4. Levin M., Pomerance C., Soundararajan K.: Fixed points for discrete logarithms. In: Algorithmic Number Theory. Lecture Notes in Computer Science, vol. 6197, pp. 6–15. Springer, Berlin (2010).
5. Menezes A., van Oorschot P., Vanstone S.: Handbook of Applied Cryptography. CRC Press, New York (1997).
6. Peirce C.: Collected Papers, Book III, vol. 4, chapter 1. Harvard University Press, Cambridge (1958).